

Graph Invariants

A graph invariant is some function f such that for graphs Γ_1 and Γ_2 ,
 $\Gamma_1 \cong \Gamma_2 \implies f(\Gamma_1) = f(\Gamma_2)$.

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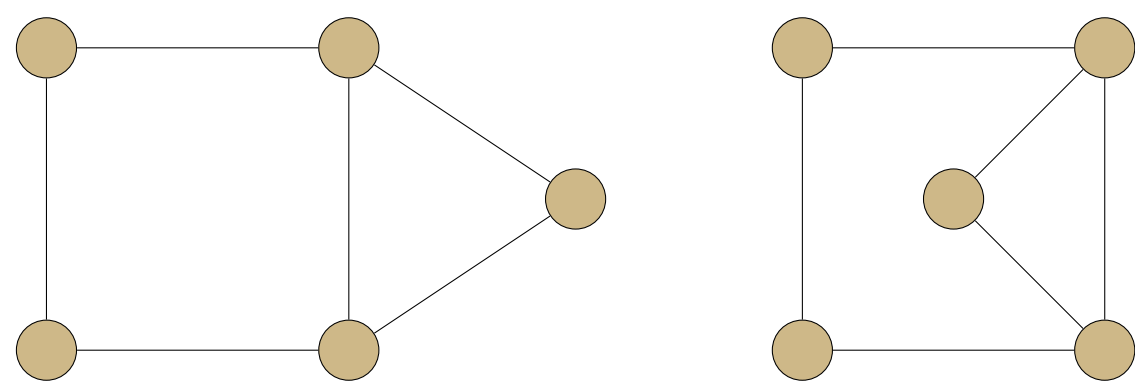


Figure 1. Pictorial representation of two isomorphic graphs

This research focuses on the *Lovász Theta* number, an invariant that bounds the NP-hard to compute chromatic and clique numbers.

$$\alpha(\overline{G}) \leq \vartheta(G) \leq \chi(\overline{G})$$

Quantum Graphs

To think about a quantum graph, it is helpful to first consider the matricial system associated to a graph:

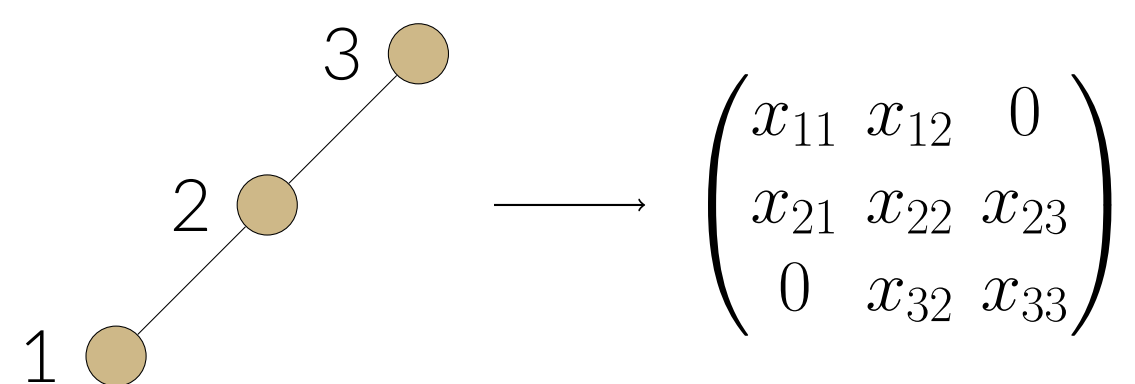


Figure 2. Example of matrix system associated to a graph

Generalizing these matricial systems, a *quantum graph* (also called a non-commutative graph) is any matricial system that is closed under taking the adjoint and contains the identity.

Quantum Graph Invariants

Consider two extensions of *Lovász Theta* for quantum graphs: the CP-index [1] and the quantum Lovász Theta [2].

$$\text{Ind}_{\text{CP}}(\mathcal{S}_1 : \mathcal{S}_2) = \inf \{ \|\varphi(I)\| : \varphi(I) \in \mathbb{C}I, \varphi(\mathcal{S}_1) \subset \mathcal{S}_2, \varphi - I \in \text{CP}(\mathcal{S}_1) \}$$

$$\tilde{\vartheta}(\mathcal{S}) = \max \{ \|I + T\| : T \in \mathcal{S}^\perp, I + T \succeq 0 \}$$

Research Question

For an arbitrary quantum graph $\mathcal{S}$, does $\text{Ind}_{\text{CP}}(M_n : \mathcal{S}) = \tilde{\vartheta}(\mathcal{S})$?

Choi Representation

Define E_{ij} to be the matrix whose i, j -th entry is 1 and whose other entries are 0. A linear map φ is fully defined by the values of $\varphi(E_{ij})$, which can be summarized in a *Choi matrix*:

$$\sum_{i,j} \varphi(E_{ij}) \otimes E_{ij}.$$

Theorem (Choi): φ is completely positive if and only if its Choi matrix is positive semidefinite.

Methods

$$\begin{aligned} & \text{maximize} && \lambda \\ & \text{subject to} && \text{tr} \otimes \text{id}(X) = (1 - \lambda)I_n \\ & && X + \lambda\Delta_n \in M_n \otimes \mathcal{S} \\ & && X \in (M_n \otimes M_n)^+ \end{aligned}$$



0.33	.	.	0.17	-0.17	-0.17	0.17	-0.17	-0.17
.	0.33	.	-0.17	0.17	-0.17	-0.17	0.17	-0.17
.	.	0.33	-0.17	-0.17	0.17	-0.17	-0.17	0.17
0.17	-0.17	-0.17	0.33	.	.	0.17	-0.17	-0.17
-0.17	0.17	-0.17	.	0.33	.	-0.17	0.17	-0.17
-0.17	-0.17	0.17	.	.	0.33	-0.17	-0.17	0.17
0.17	-0.17	-0.17	0.17	-0.17	-0.17	0.33	.	.
-0.17	0.17	-0.17	-0.17	0.17	-0.17	.	0.33	.
-0.17	-0.17	0.17	-0.17	-0.17	0.17	.	.	0.33

Figure 3. Example Choi Matrix output from the semidefinite program



$$\varphi(x) = \frac{X \bullet J}{n} I - \frac{X \bullet (J - I)}{n(n-1)} J$$

Result

Defining $\mathcal{T}_A = (\text{span}\{A\})^\perp$, the following are true:

$$\begin{aligned} \text{Ind}_{\text{CP}}(M_n : \mathcal{T}_{J-I}) &= 2, \\ \tilde{\vartheta}(\mathcal{T}_{J-I}) &= n. \end{aligned}$$

Thus the two Quantum Lovász Theta numbers do not agree in general.

Further Conjectures

$$\begin{aligned} \text{Ind}_{\text{CP}}(M_n : \mathcal{T}_A) &= 2, \\ \text{Ind}_{\text{CP}}(M_n : \mathcal{T}_A^*) &= n, \\ \text{Ind}_{\text{CP}}(\mathcal{T}_A^* : \mathbb{C}I) &= n, \end{aligned}$$

Given two matrix systems of graphs Γ_1, Γ_2 of size $n \times n$, define the “unitary perturbation” by a unitary matrix U from the Haar distribution by $\Gamma_1 \sigma \Gamma_2 := U \mathcal{S}_{\Gamma_1} U^* + \mathcal{S}_{\Gamma_2}$. Hypothesis:

$$\text{Ind}_{\text{CP}}(M_n : \Gamma_1 \sigma \Gamma_2) \leq \min \{ \text{Ind}_{\text{CP}}(M_n : \mathcal{S}_{\Gamma_1}), \text{Ind}_{\text{CP}}(M_n : \mathcal{S}_{\Gamma_2}) \}.$$

One may go a step further in this direction and compute the average inequality over a number of unitary matrices from such Haar distribution as and make the following conjecture:

$$\mathbb{E}[\text{Ind}_{\text{CP}}(M_n : \Gamma_1 \sigma \Gamma_2)] = \mathbb{E}[\tilde{\vartheta}(\Gamma_1 \sigma \Gamma_2)],$$

where the the expected value is over the aforementioned set of unitary matrices.

Acknowledgments

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References

- [1] R. Araiza, C. Griffin, and T. Sinclair. An index for inclusions of operator systems, 2022.
- [2] R. Duan, S. Severini, and A. Winter. Zero-error communication via quantum channels, noncommutative graphs, and a quantum lovász number. *IEEE Transactions on Information Theory*, 59(2):1164–1174, Feb. 2013.